

NEPHELE

Business Model Document

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1. Executive Summary

Nephele is aiming to restore the marine cloud brightening effect that shipping produced for over a century and was abruptly removed when the IMO 2020 sulfur cap took effect. A bolt-on module installed above the exhaust stack of an existing scrubber-equipped vessel injects atomised seawater into the warm, water-saturated post-scrubber plume. The plume delivers the droplets to the cloud base as pre-activated cloud condensation nuclei, using the ship's own energy and existing seawater circuit. Over 4,000 scrubber-equipped vessels are in service today; none require new construction.

The technical case is set out in the Nephele whitepaper (available at nephele.earth). The commercial case rests on three points. First, IMO 2020 opened a measurable warming gap (+0.06 to +0.2 W/m²) that the IMO is under increasing pressure to address. Second, the IMO Net-Zero Framework (NZF), currently in development under MARPOL Annex VI, provides the financial disbursement vehicle: Regulation 39.1's "notwithstanding" clause gives the MEPC explicit discretionary authority to recognise ZNZ technologies outside the standard fuel-intensity metric, and the guidelines defining that pathway are being developed imminently. Third, Nephele onboard emission monitoring is a standalone compliance verification instrument for the existing scrubber fleet, generating revenue independently of ship track restoration.

The company is at a very early stage. Nephele is pre-incorporation (anticipating UK incorporation). Neither co-founder is full-time and no capital has been raised. Both founders are engaged with the venture in parallel with existing employment, which has removed pressure to raise dilutive capital prematurely; however, both are willing to depart from their current positions given favorable circumstances around the venture.

The ask: £1.4M for Phase 0 - an 18-month measurement, hardware development, and regulatory engagement program - with a strong preference for non-dilutive sources (ARIA funding, Innovate UK, climate philanthropy such as Bezos Earth Fund). Phase 1 (pilot, 12-18 months) requires an additional £5.0M assembled from continued grants, philanthropy, shipping-partner capital expenditure, and in-kind vessel access. From Phase 2 onward the business should be self-funded through NZF disbursements and emissions verification revenue.

2. Problem and Regulatory Context

The IMO 2020 cooling gap

For over a century, sulfur dioxide from marine fuel oxidized in the atmosphere to form sulfate aerosol that seeded brighter, more reflective marine clouds. The effect was visible from space as linear "ship tracks" behind vessels in every major ocean corridor (illustrative image available on Nephele website homepage). In January 2020, the IMO sulfur cap cut shipping sulfur emissions by approximately 80% - a substantial improvement for public health in port cities, and an unambiguously correct decision on its own terms.

The unintended consequence was to remove most of that cloud-brightening effect. The measured radiative forcing from the reduction is +0.06 to +0.2 W/m² globally, equivalent to one to three years of

additional greenhouse warming arriving all at once (Yuan et al. 2024; Jordan et al. 2024; Gettelman et al. 2024). Hansen et al. (2026) argue this is one component of a broader decline in aerosol cooling driving the sea surface temperature acceleration observed since 2023. Diamond and Boss (2025) used the Cape of Good Hope shipping diversion as a natural experiment and quantified the cloud response reduction at approximately 67% per unit of shipping.

The technical detail is set out in the Nephele whitepaper. What matters commercially is that the gap is measured, attributable to a specific IMO regulation, and increasingly recognized as a problem the IMO itself is responsible for addressing.

The regulatory opening

The IMO Net-Zero Framework was approved in principle at MEPC 83 in April 2025. Formal adoption was adjourned at the extraordinary session in October 2025 by a contested vote of 57 to 49, with opposition led by the United States, China, Russia, Saudi Arabia, Singapore, and Liberia. Adoption is currently expected at the next extraordinary session; the framework's earliest entry into force is March 2028, with fund disbursements beginning no earlier than 2029, allowing a short window for Nephele to take action.

Regulation 39.1 of draft revised MARPOL Annex VI defines the standard ZNZ threshold as a well-to-wake GHG fuel intensity $\leq 19.0 \text{ gCO}_2\text{eq/MJ}$ (tightening to 14.0 by 2035). Sea salt aerosol injection consumes no fuel and produces no GHG emissions, so the standard metric does not apply (meaning Nephele will need to generate a separate measurement for the impact created). The critical provision is the final sentence: "Notwithstanding, the Committee may approve additional ZNZs taking into account guidelines developed by the Organization." This grants the MEPC explicit discretionary authority to recognise ZNZ technologies outside the standard fuel-intensity framework. Those guidelines - expected ahead of or at MEPC 85 in late 2026 - are the specific regulatory window Nephele is engaging with. See whitepaper Sections 12.1-12.4 for the full legal analysis.

Adoption risk

The MEPC 85 vote is a material risk. If NZF adoption stalls, is materially watered down, or is delayed significantly, the timeline to disbursement extends and the guideline drafting window Nephele is working within is disrupted (though may end up benefitting Nephele's continued development). This risk is addressed structurally in Sections 5 and 10: the emission verification revenue stream and voluntary corporate climate programmes are alternate paths that do not depend on NZF adoption. Nephele's value is not solely a bet on the fund.

3. Solution and Technology

The full technical treatment is in the whitepaper. What follows is the commercial-audience summary.

The Nephele module is a bolt-on unit mounted above the existing scrubber exhaust stack, above the scrubber's own mist eliminator. It does not modify the scrubber or affect its MARPOL Annex VI compliance. Three components: air-assist spray nozzles feeding a patent-pending rotary splash atomiser, a replaceable mist eliminator cassette above it, and instrumentation and control electronics. Seawater is drawn from the existing scrubber seawater circuit; no additional intake, filtration, or power source is required.

The core insight is that the post-scrubber exhaust plume is the most efficient CCN delivery mechanism available. The plume exits the stack at higher than ambient temperature and 100% relative humidity, rising buoyantly at multiple meters per second. Marine boundary layer humidity in stratocumulus regions exceeds the NaCl deliquescence point (~75% RH) for 85-95% of the time (Bretherton et al. 2010; Cadeddu et al. 2023), which means injected droplets remain in solution throughout transit and arrive at cloud base as pre-activated CCN - no drying step, no deliquescence barrier, no activation delay. The whitepaper hypothesises 70–95% end-to-end delivery efficiency; ambient (sea-level spray) approaches are bounded at 0.5–10.6% (Hernandez-Jaramillo et al. 2023, 2025).

The patent-pending element is the rotary splash atomiser, which solves the film-thickness problem that suppresses splash atomisation on any static plate at commercial throughput. A rotor spinning at ~6,000 RPM substitutes centrifugal acceleration (~4,000 g) for gravitational drainage, thinning the equilibrium film from ~180 µm to ~2.8 µm and pushing the daughter droplet distribution into the sub-3 µm range which is useful for both passing through the mist eliminator and catalyzing cloud formation. The result is approximately 30x more useful CCN per litre of injected seawater than off-the-shelf air-assist nozzles alone, without the significant cost of these specialty nozzles.

Delivery efficiency figures and per-ship CCN output are hypotheses. They will be measured directly in Phase 0. The document does not assert them as validated.

Off-switch. Sub-micron sea salt aerosol has an atmospheric residence time of hours to days. If injection stops, cloud brightening ceases within 24-48 hours without persistent atmospheric modification. Cessation authority is unconditional and retained by the MEPC, flag states, and port states. This is a material differentiator from any other climate intervention proposal and is central to the regulatory framing.

4. Regulatory Pathway and Revenue Mechanism

The pathway

Three regulatory outcomes would qualify Nephele for NZF disbursements, in order of preference:

- **Option A (preferred):** The ZNZ guidelines developed under Regulation 39.1 include an explicit pathway for ship-installed technologies producing verified net negative radiative forcing, with the W/m²-to-CO₂eq equivalence methodology defined as part of the same guidelines package
- **Option B:** The guidelines define ZNZ eligibility broadly enough - consistent with the "technologies" language already in Regulation 39.1 - that any ship-installed technology producing verified net climate benefit qualifies. Nephele would seek formal recognition through the standard guidelines application process
- **Option C (fallback):** A separate disbursement category under the Fund's climate protection and environmental resilience mandate (Regulation 40), distinct from the standard ZNZ mechanism but drawing from the same Fund

None of these options require amending the settled MARPOL Annex VI text; the guidelines are still under development. The critical window is the drafting period ahead of MEPC 85 in late 2026. Nephele's regulatory work is directed at ensuring the guidelines include a workable pathway rather than at forcing a bespoke recognition.

The revenue mechanism

Once the framework is operational (2029 earliest), the payment flow should be straightforward, assuming the NZF comes into effect as expected. Verified radiative forcing from each participating vessel is measured via the satellite attribution pipeline (whitepaper Section 9.2), converted to CO₂eq avoided using the approved methodology, multiplied by the reward rate set by the Committee, and disbursed by the Fund on an annual reporting cycle. Nephele invoices the Fund and remits the contracted margin to partnered shipping operators as compensation for carrying the equipment, helping to offset their GHG compliance costs, and retains the remainder.

Two parameters remain to be set through the guidelines process:

- **The reward rate** (per tonne CO₂eq), to be defined by the Committee no later than 1 March 2027 under Regulation 39.3.
- **The W/m²-to-CO₂eq conversion methodology**, which does not yet formally exist. Developing this equivalence is a core objective of Nephele's MEPC engagement; the science is grounded in the same peer-reviewed literature that quantifies the IMO 2020 warming effect.

Per-vessel revenue projections cannot be made honestly until both parameters are set. Section 5 gives the structural framework and a scenario sensitivity in the interim.

5. Business and Revenue Model

Two revenue streams, sequenced by time horizon

Nephele has two independent revenue streams. Their sequencing is important: the near-term stream begins from Phase 1 and does not depend on NZF adoption; the long-term stream begins from 2029 at the earliest and depends on the regulatory outcomes discussed in Section 4.

Stream	Emission verification (near-term)	NZF disbursements (long-term)
First revenue	Phase 1 (2028), from first paying subscriber	2029 earliest, dependent on NZF adoption
Customer	Scrubber-equipped vessel operators, classification societies, flag-state authorities	IMO Net-Zero Fund (Nephele invoices; shares with partnered operators)
Depends on NZF adoption?	No	Yes
Addressable market	4,000+ scrubber-equipped vessels operating today	The same fleet retrofitted with Nephele modules
Key sensitivities	Subscription price, customer acquisition and churn, penetration rate, MARPOL enforcement rigour	Reward rate, W/m ² -to-CO ₂ eq conversion, cloud response per vessel

5.1 Emission verification revenue (near-term)

The Nephele mist eliminator cassette inherently carries a tamper-resistant chemical record of both scrubber and salt injector operation: total NaCl mass quantifies cumulative seawater throughput, sulfate-to-sodium ratio in the deposits gives an independent, time-integrated measurement of exhaust SO₂ concentration, and total deposit mass constrains operating hours (whitepaper Sections 9.1, 13.3).

MARPOL Annex VI requires scrubber-equipped vessels to demonstrate compliance with sulfur emission limits, but current verification relies primarily on onboard electronic logging with periodic port state control inspections - a system with documented monitoring gaps. The cassette is a physical,

chemistry-based, tamper-evident verification record that is independent of onboard electronics. It has standalone value to classification societies, flag states, and the IMO for the existing 4,000+ scrubber fleet, entirely independent of ship track restoration.

Aside from the unique physical deposition component of the Nephele emissions monitoring package, environmental instrumentation will already be required for accurate salt injection control. Stacking CH₄, CO₂, or other GHG sensors on top of our baseline sensor package would be fairly trivial for continuous GHG emissions monitoring.

Commercial model. Subscription supply-and-retrieval: Nephele manufactures and supplies tamper-resistant emissions monitoring modules to scrubber operators on a defined swap cycle (target timing: aligned with routine port-call schedules), retrieves and analyses them centrally, and provides verified emissions records to operators, classification societies, or flag-state authorities as required. Revenue begins from the moment emissions monitoring is in operation - before NZF disbursements. Pricing, cycle length, and analytical margin are to be developed with pilot partners; there is no confirmed customer or pricing point today.

Illustrative sizing. *The following figures are illustrative and not projections.* At full addressable market (~4,000 vessels), a subscription price of \$10–30k per vessel per year would imply a global TAM of \$40–120M annually for this stream alone. Penetration is unlikely to be linear; scaling this stream will depend on securing at least one classification society or flag-state endorsement of the system as an acceptable compliance verification record. That endorsement is a Phase 0-1 objective.

5.2 NZF disbursements (long-term)

The IMO Net-Zero Framework, once operational, will collect revenue through mandatory GHG pricing contributions from non-compliant vessels and disburse those funds as rewards to ships operating ZNZ technologies. The Fund is projected to generate \$10–12 billion annually in its initial years (TESS Forum 2025). Nephele's share is a function of the fleet it equips, the verified radiative forcing per vessel, the reward rate, and the W/m²-to-CO₂eq conversion.

Structural framework. Once the reward rate and conversion methodology are set, revenue for a fleet of N vessels is approximately:

$$\text{Revenue} \approx N \times RF_{\text{vessel}} \times f \times R$$

where RF_{vessel} is the annualised verified radiative forcing per vessel (W/m² over the vessel's operating footprint), f is the approved W/m²-to-tCO₂eq conversion factor, and R is the reward rate (\$/tCO₂eq).

Nephele's share of that revenue is set by the operator contract - the balance is remitted to the partnered shipping operator as compensation for carrying the equipment and offsetting their own GHG compliance costs under the same framework.

Illustrative scenario. *The following is a scenario for orientation, not a projection.* If R settles in the range comparable to current voluntary carbon markets or the EU ETS (\$50–200/tCO₂eq), and the eventual W/m²-to-tCO₂eq conversion produces per-vessel annual CO₂eq equivalents in a plausible mid-range consistent with published ship-track literature, the revenue per equipped vessel is in the same order of magnitude as a scrubber's own annual fuel-cost benefit. That is enough to make the capex-financing model in Section 5.3 workable, but it is not a claim about specific dollars. The number will be replaced

with a defensible projection as soon as (a) the reward rate is set at MEPC and (b) the Phase 0 measurement gives a defensible RF-per-vessel figure.

5.3 Per-vessel unit economics and financing

Estimated per-vessel installation cost is approximately £200,000 including hardware, installation, and commissioning (whitepaper Section 5.5). For context, a scrubber itself costs \$2-8M to install. The Nephele module is removable, module-based, and independent of the vessel's existing financing.

The scrubber industry established the relevant precedent ahead of IMO 2020: capex-sharing clauses, profit-share arrangements, and third-party leasing structures emerged as the payback calculus became clear (HFW briefing on scrubber financing structures; Norton Rose Fulbright and Value Maritime are also citable). The natural structure for Nephele is analogous: the £200k per-vessel upfront is financed by the operator against recurring fund disbursements, with Nephele's share taken as a service margin from the disbursement flow rather than as an equipment sale. This aligns incentives (Nephele is paid only when the module produces verified forcing) and preserves operator cash flow (no large capital outlay against uncertain future revenue).

Cost structure. Nephele's cost base is largely fixed once the system is developed and verification infrastructure is built. The satellite attribution pipeline, data registry, and central analytical instrumentation are one-time builds that scale across the fleet at low marginal cost. Variable per-vessel costs are (i) emissions monitoring module manufacture and analysis, (ii) periodic shipboard instrumentation servicing, and (iii) a share of ongoing regulatory and platform-engineering costs. Gross margin at scale is therefore driven by fleet size against a largely-fixed overhead base rather than by per-vessel economics improving. This distinction matters: the framing "margins improve with scale" refers to overhead absorption, not to per-vessel cost efficiency.

6. Market and Go-to-Market

Market size and segmentation

The addressable retrofit market is the existing global scrubber-equipped fleet: approximately 4,000+ vessels as of 2026 (industry trackers put the number at 4,000 to 5,000 depending on definition). No new vessel construction is required. Priority corridors are those with high marine stratocumulus prevalence - Northeast Pacific, Southeast Atlantic, Southern Ocean - where cloud brightening produces the largest per-particle radiative forcing.

Customer segmentation:

- **Shipping majors** (Maersk, Hapag-Lloyd, MSC, CMA CGM) - potential pilot partners in Phase 1, eventual fleet customers from Phase 2 onward. Selection criteria: scrubber fleet size, corridor overlap with stratocumulus regions, and public ESG commitments.
- **Classification societies and flag states** - customers for the emission verification service, and regulatory validators for the compliance-record framework. Approach is coordinated with Nephele's IMO engagement.
- **IMO / MEPC / ISWG-GHG** - not a customer in the revenue sense, but the party whose recognition of the ZNZ pathway underwrites the long-term revenue stream. Institutional engagement here is a first-order commercial activity, not a policy overhead.

Go-to-market motion

Current motion is warm-introduction driven and directly reflects the pre-incorporation stage. Priority channels:

- **Maersk** - Maersk Growth, an internal team focused on ESG ventures, via a mutual contact
- **ARIA (Exploring Climate Cooling programme)** - UK government advanced research funder; strongest single near-term fit for Phase 0
- **Climate philanthropy** - Bezos Earth Fund, Grantham Foundation, Breakthrough Energy, CIFF, Schmidt Futures

A cold enterprise sales process into shipping majors is deliberately deferred until Nephele has (a) formal incorporation, (b) at least one signed pilot letter of intent, and (c) an ARIA or equivalent grant to point to. Warm introductions are the credible motion at this stage.

NZF duration and long-term market dynamics

Open question flagged for further work with IMO counsel: the operational duration of the Net-Zero Fund itself. Current framework materials describe the fund as ongoing but the specific review, sunset, or reauthorisation cycle is not yet defined in publicly available text. If the fund operates on a defined-term basis with periodic reauthorisation, that becomes a structural risk for Phase 2 revenue durability and needs to be addressed in the operator contract terms - likely via a clause tying operator remittance to actual disbursement receipts rather than to a fixed period. This is a documentation gap to close, not a business-model problem.

7. Competitive Landscape

The marine cloud brightening field is scientifically well-established (Twomey 1977; ship tracks documented since 1966) but has produced very few deployed systems. Existing efforts split into three categories:

- **Ambient (sea-level) spray proposals.** Salter's original micro-fabricated silicon nozzle concept; the Cambridge Centre for Climate Repair's converging-diverging nozzle work; the University of Washington effervescent nozzle programme (Harrison et al. 2025) whose Alameda coastal trial was suspended in 2024 following community engagement issues. All rely on spraying seawater at sea level, drying to dry salt crystals, and passive marine boundary layer turbulence to loft the crystals to cloud base. Delivery efficiency for this pathway is bounded at 0.5–10.6% end-to-end (Hernandez-Jaramillo et al. 2023, 2025).
- **Research programmes.** The Australian Reef Restoration and Adaptation Program (RRAP) has run localised MCB research targeted at Great Barrier Reef cooling. It is research-stage rather than commercial. Silver Lining and adjacent US-based non-profits do policy, research, and advocacy work on solar radiation modification broadly and are not direct commercial competitors.
- **Onshore / land-based experiments.** Small-scale outdoor experiments at various universities have not produced deployment-ready systems.

Nephele's differentiation is not the hardware. Nozzles, atomisers, and mist eliminators are off-the-shelf or lightly modified industrial components. The rotary splash atomiser is patent-pending and non-trivially defensible, but a determined competitor could plausibly design around it. The moat is elsewhere:

- The verification stack - cassette chemistry, shipboard instrumentation, satellite attribution pipeline, and the statistical transfer function built during Phase 1 - is what makes the emissions and cloud-response claims auditable to the IMO. That combination has no current competitor.
- The institutional relationship development with the MEPC and the operational track record of having helped write the guidelines under which the technology is recognized. Trusted-operator status is not something a later entrant can retroactively acquire.

- The installed base of pilot vessels before Fund disbursements begin. A partner who has run a Nephele module for two years before 2029 is not a prospect - they are a paying customer waiting for the mechanism to activate.

The Nephele vs. ambient MCB comparison from the whitepaper (Section 11):

Parameter	Ambient MCB (sea-level)	Nephele (plume-delivered)
Vessel fleet	Purpose-built spray ships	4,000+ existing scrubber vessels
Delivery efficiency	0.5 – 10.6% (published)	70 – 95% (hypothesis, subject to Phase 0 test)
Per-unit cost	\$10M+ per unit	~\$250k per module
Deployment timeline	Years to build out	Months to retrofit
Regulatory pathway	Undefined	IMO MEPC oversight pathway defined

8. Team and Structure

Nephele is pre-incorporation as of August 2026. Nephele Ltd (UK) has been confirmed available on Companies House and is the intended incorporation vehicle. Zane will incorporate as sole initial director; Creighton will be added via co-founder agreement. December year-end is selected for the accounting period. No capital has been raised. Both founders currently hold other full-time employment; neither is drawing salary from Nephele. Titles used externally (CEO, CFO) are aspirational and reflect anticipated post-funding roles.

Co-founders

Zane Grissett - Science and Hardware Lead. BS in geology & geological oceanography, University of Rhode Island. Robotics and test engineering experience at Ample and Google X. Hardware and commissioning lead for superconducting magnet test infrastructure at Commonwealth Fusion Systems. Owns the technical and scientific work at Nephele: whitepaper physics, hardware design and prototyping, testing and measurement campaign design.

Creighton Kling - Commercial and Regulatory Lead. BBA Finance, University of Notre Dame. Vice President in investment banking at Deutsche Bank, with M&A, equity and debt capital markets experience across large-cap technology and industrials transactions in Chicago, San Francisco, and London. London-based, which is operationally material given the IMO's London headquarters. Owns commercial strategy, investor and grant narrative, shipping-partner relationships, IMO stakeholder engagement, financial architecture, and the business documentation set.

9. Milestones, Phases, and Use of Funds

Three phases. Each is triggered by the completion of the prior phase's deliverables and the corresponding funding tranche.

Phase 0 - Measure, build, engage • 18 months • £1.4M

Three parallel tracks: (A) a plume measurement campaign to quantify CCN concentration, size distribution, RH profile, and dispersion in a post-2020 scrubber-equipped ship plume - data that does not exist in peer-reviewed literature; (B) bench-scale hardware development of the rotary splash atomiser, mist

eliminator with seawater spray, and emissions monitoring module development; (C) IMO MEPC delegate engagement and technical input to the ZNZ guidelines, including a first draft of the W/m²-to-CO₂eq conversion methodology.

Phase 0 line item	Budget
Personnel - 2 FTE × 18 months (founders)	£400k
Hardware development and lab build-out (rotor prototypes, nozzle rigs, seawater loops, space)	£250k
Analytic equipment (CCN counter, SMPS/APS, high-speed imaging), external testing/validation	£250k
Plume measurement campaign (vessel access, travel, instrumented drone, mobile lab equipment)	£250k
Regulatory engagement (MEPC travel, IMO delegation liaison)	£100k
Legal and IP (PCT filing, UK Ltd formation, pilot agreements)	£80k
Admin, insurance, overhead, contingency (~10%)	£70k
Phase 0 total	£1,400k

Key milestones. M6 - bench-scale rotor prototype v1; vessel partner secured. M9 - first plume campaign dataset in hand. M12 - Regulatory engagement, MEPC input M18 - peer-reviewed paper submitted; PCT filing from provisional; Phase 1 go/no-go.

Phase 1 - Three-vessel pilot · 24 months · £5.0M

Modules installed on three scrubber-equipped cargo vessels operating regular routes through marine stratocumulus corridors. Two objectives: (i) measure whether Nephele's CCN flux produces a detectable cloud response - the central empirical test; (ii) build and calibrate the verification transfer function (shipboard instrumentation predictors → satellite/aircraft-measured cloud response) that enables fleet-scale verification without aircraft tailing every vessel.

Phase 1 line item	Budget
Module fabrication - 3 vessels (~£250k per vessel: hardware, rotor, monitoring, controls)	£900k
Vessel integration and commissioning (installation at port calls, scrubber interface, HSE)	£400k
Shipboard Tier-1 instrumentation (OPC, CCN-C, ceilometer, weather station, telemetry)	£200k
Tier-2 research aircraft campaigns (×2) - 2–4 week deployments, two regions, two seasons	£900k
Expanded team - 8 FTE × 24 months (eng, atmos science, data, regulatory, ops)	£2,200k
Satellite data and analysis (CERES/MODIS pipeline, counterfactual attribution)	£200k
Regulatory engagement (continued IMO, flag-state, classification society)	£150k
Admin, overhead, contingency	£50k
Phase 1 total	£5,000k

Key milestones. M6 - Vessel 1 injecting. M12 - first cloud-response dataset. M18 - verification transfer function validated. M24 - three vessels operational; fleet-scale verification framework ready.

Phase 2 - Scale · 2030 onward · self-funded

From 2029 onward, NZF disbursements (assuming NZF adoption per Section 4) plus emissions monitoring subscription revenue are expected to fund fleet expansion, additional vessel retrofits, and continued regulatory engagement. Phase 2 is the point at which Nephele transitions from grant-funded R&D and pilot operations to self-sustaining commercial operations.

10. Risks and Mitigants

Category	Risk	Mitigant
Regulatory / political	NZF adoption stalls, is watered down, or is delayed materially beyond current timelines. October 2025 vote was contested (57–49); active opposition from US, China, Russia, Saudi Arabia, Singapore, Liberia.	Emissions verification revenue stream and voluntary corporate climate programmes are alternate paths that do not depend on NZF adoption. Nephele's commercial case is not solely a bet on the fund. Continued MEPC engagement supports the specific guideline outcomes needed regardless of overall NZF adoption pace.
Technical	Plume delivery efficiency (70–95% hypothesis) and per-ship CCN flux are not yet directly measured. Rotary splash atomiser unproven in continuous marine operation. Cloud response nonlinearity - adding CCN to an already-perturbed plume produces less brightening per particle.	Phase 0 measurement campaign is designed to test these hypotheses directly and is the primary use of Phase 0 capital. Falsifiable, staged plan with explicit go/no-go at M18. Bench-scale test rig already scoped; nozzle-agnostic platform allows technology substitution if the rotary atomiser underperforms.
Operational / safety	Unintended cloud darkening from drizzle induction. Local ecological or health effects from injected aerosol. Interference with scrubber MARPOL compliance.	48-hour atmospheric reversibility (sea salt residence time in the MBL). Hard-stop conditions defined in advance (whitepaper Section 14.2): net cloud darkening over 10+ transits; drizzle >2× baseline over 5+ transits; particles >3µm passing eliminator >5% by number; MARPOL interference; MEPC/flag-state/port-state directive. All results publicly published.
Commercial	No signed shipping partner as of July 2026. Warm outreach in progress but no LOI. Reliance on ARIA and philanthropic grant timelines for Phase 0.	Multiple parallel outreach threads (Maersk venture team, Maersk strategic innovation, Ishoda, dorm-founder introduction, whisky-club climate investor). Grant applications sequenced to overlap. Founders remain in existing employment until funding conversion, removing personal financial pressure.
Reputational	"Geoengineering backlash" - public perception that any climate intervention is unacceptable or destabilising. University of Washington coastal MCB trial suspended in 2024 following community engagement issues is a cautionary precedent.	"Restoration not engineering" framing (returning a pre-existing forcing, not initiating a novel one). Open-ocean-only operating envelope, programmatic deactivation in coastal and territorial waters. MEPC oversight framing throughout. Open data commitments. 48-hour off-switch as central talking point.

11. The Ask

Phase 0 - £1.4M. 18-month programme covering the measurement campaign, hardware development, and IMO regulatory engagement (Section 9). Strong preference for non-dilutive sources: ARIA Exploring Climate Cooling as the primary target, followed by Innovate UK, Bezos Earth Fund, Grantham Foundation, Breakthrough Energy, ClIFF, and Schmidt Futures.

Phase 1 - £5.0M. 24-month pilot including the verification architecture build (Section 9). Blended stack: continued grants and philanthropy, shipping-partner CAPEX on module installation, in-kind vessel access, and - if the regulatory pathway is on track by that point - mission-aligned equity from climate-tech investors comfortable with pre-revenue regulatory-dependent business models.

Phase 2 - self-funded. NZF disbursements from 2029 plus emissions verification subscription revenue.

What changes with funding. The Phase 0 grant is the trigger for full-time commitment: both founders leave existing employment; Nephele Ltd is fully staffed on the schedule in Section 9; the co-founder agreement is executed; the first Phase 0 hires begin. Between now and that point, work continues in parallel with existing employment on the current cadence - pitch materials, warm outreach, whitepaper iteration, and pre-incorporation setup.

12. Appendix

A. Acronym glossary

ARIA - UK Advanced Research and Invention Agency

CCN - Cloud Condensation Nuclei

GFI - Greenhouse Gas Fuel Intensity (well-to-wake, gCO₂eq/MJ)

HFO - Heavy Fuel Oil

IMO - International Maritime Organization

ISWG-GHG - IMO Intersessional Working Group on Reduction of GHG Emissions from Ships

LOI - Letter of Intent

MARPOL Annex VI - International Convention for the Prevention of Pollution from Ships, Annex VI (air pollution)

MBL - Marine Boundary Layer

MCB - Marine Cloud Brightening

MEPC - IMO Marine Environment Protection Committee

NZF - IMO Net-Zero Framework / Net-Zero Fund

Reg 39.1 - Regulation 39.1 of draft revised MARPOL Annex VI - the discretionary "notwithstanding" clause on which ZNZ recognition for Nephele depends

RF - Radiative Forcing (W/m²)

SO₂ - Sulfur dioxide

TAM - Total Addressable Market

ZNZ - Zero or Near-Zero (emission technology)

B. Source materials

- Nephele whitepaper V6.1, April 2026 (nephele.earth) - technical foundation, particle physics, verification architecture, regulatory analysis, and full reference list.

- Nephele pilot programme proposal, April 2026 (v2) - 22-slide deck; Phase 0/1 budget detail and milestone framing.
- IMO document MEPC/ES.2/2 - draft revised MARPOL Annex VI (125 pages); ZNZ framework, penalty structure, fund mechanics, disbursement provisions.
- Diamond & Boss (2025) - 67% cloud response reduction per unit of shipping from IMO 2020 sulfur cap.
- Hansen, Kharecha, Morgan & Vest (2026) - climate sensitivity and reduced-aerosol-cooling arguments for accelerated SST warming.
- HFW, Norton Rose Fulbright, Value Maritime - citable sources for post-IMO 2020 scrubber CAPEX-sharing and third-party leasing financing structures (basis for the operator financing model in Section 5.3).
- TESS Forum (2025) - Equity in Motion: Revenue Disbursement Under the IMO Net-Zero Framework; \$10–15B annual fund size projection.

D. Open questions / follow-ups

- NZF operational duration - the fund is expected to be reviewed and renewed after 5 years, but that is not 100% confirmed
- NZF qualification, payout proportions, etc. - how is the fund paid out? When? What portion of the fund will be earmarked for what purposes? These questions may not be answered until the legislation is finalized at a later date
- W/m²-to-CO₂eq conversion - draft first-pass methodology for Nephele's MEPC submission, potentially based on early-stage testing or modeling; align framing with published aerosol-forcing literature
- Cassette pricing and analytical margin - gather information on availability, usage, pricing and margin for emissions measurement cassettes
- Immigration / visa position - Creighton is located in the UK on a work visa; taking on substantive full-time Nephele work (i.e. terminating current employment) will likely void his current visa; pursue alternative visa sponsorship or relocate back to the US